SunMint Tree-Growth Monitoring — Video Capture + Python Analysis Worker

Spec v1.1 · 2026-08-25 · Prepared by Sophia Truesight (autopilot) for Gary Teh

Scope: monitor_tree_growth module (DApp + SunMint app video capture) + backend Python analysis worker + PM002 carbon math + on-chain event flow.

1. WHY

The SunMint model requires per-tree growth evidence that a VVB can audit. Phone-camera measurement is the proven pattern (TREEO 94–95% DBH accuracy $R^2 \geq 0.95$; ACORN/Agerpoint moving ground-truth to phone; Greenstand photo-verified). Video adds stronger tamper-evidence than a single photo — the walk-around cadence (farm-build game) is captured continuously.

2. SYSTEM ARCHITECTURE

```

Farmer phone (SunMint app + main DApp monitor page) Backend (Python worker)

■■ record video: walk-around tree, calibration card visible ■■ [ingest] receive video + metadata

■■ auto-extract GPS + timestamp + farmer RSA signature ■■ [hash] SHA-256 of video → audit trail

■■ save draft locally (offline-first) ■■ [process] OpenCV:

■■ POST signed metadata + video → /api/monitor/ingest ■ frame extraction (e.g. 1 fps)

■ calibration-card detection (ARUCO/contour)

■ DBH estimation (pixel ratio → trig)

■ species lookup → allometric formula

■ → AGB → BGB (R=0.32) → ×0.47 → ×44/12

■■ [emit] signed [TREE GROWTH MONITORING EVENT] → Edgar

■■ [notify] farmer growth result in app
```

Key decisions

- Capture + lightweight math in browser; heavy CV in Python. GAS is glue only (relay/notify), NOT the analysis engine (no OpenCV runtime).
- Hash the video before processing → reproducible, tamper-evident pipeline (VVB-grade).
- Offline-first — farmers in the field record, queue, upload when online (SunMint mobile is already offline-first).
- **Dual deployment:** the monitor page ships to BOTH the main DApp (dapp.truesight.me) AND the SunMint app (sunmint.truesight.me).

3. MODULE A — monitor_tree_growth (DApp + SunMint app)

3.1 Page: monitor_tree_growth.html — deployed to BOTH DApp targets

- **Main DApp:** dapp_beta/monitor_tree_growth.html → dapp.truesight.me/monitor_tree_growth.html
- **SunMint app:** truesight_me_beta/sunmint/monitor-tree-growth/index.html → sunmint.truesight.me/monitor-tree-growth/
- Same codebase/behavior in both; crypto + offline queue identical (WebCrypto RSA-2048 keypair in localStorage, IndexedDB draft queue)

- Select farm → select/create tree record (per-tree ID, species, planting date)
- Video capture: `<input type="file" accept="video/*" capture="environment">` + live preview via getUserMedia (existing pattern in report_tree_planting.html)
- Recording guidance overlay: hold card against tree at 1.3 m (breast height), portrait, walk 360° slowly
- Auto-metadata: GPS, timestamp, farmer public key
- Offline queue: IndexedDB draft storage; upload on reconnect
- Signed POST: farmer RSA signature over metadata + video hash

3.2 Data contract (request)

```
{ farmer_id, tree_id, species, video_b64, video_sha256,
gps {lat,lng}, captured_at, farmer_signature }
```

3.3 Data contract (response)

```
{ status, measurement_id, dbh_cm, height_m, agb_kg, co2e_kg,
confidence, analysis_sha256, edgar_event }
```

4. MODULE B — Python analysis worker (truesight_autopilot)

4.1 Stack

- Python 3.11+, OpenCV (frame extraction, ARUCO card detection), numpy, hashlib
- Optional: scikit-image / lightweight YOLO for tree segmentation (v2 polish)
- Crypto: farmer public key verification (RSA-2048), result signing (Sophia DAO key)
- Deployment: on the autopilot box (or a small EC2 worker) exposed as /api/monitor/ingest

4.2 Pipeline (pure functions, unit-testable)

1. extract_frames(video, fps=1) → frames[]
2. detect_calibration_card(frame) → card corners (ARUCO marker or contour; ISO 7810 85.6×53.98 mm)
3. estimate_dbh(frame, card_bounds, px_to_mm) → DBH at 1.3 m (similar-triangle / pixel-ratio)
4. allometric_agb(species, dbh) → AGB kg (species table; IPCC/Chave allometrics)
5. carbon_math(agb) → PM002 chain: ΔBGB = ΔAGB × R (R=0.32) · CO■e = ((ΔAGB+ΔBGB) × 0.47) × 44/12 × (1–A_pre)(1–A_unc)(1–LD)(1–AR)(1–RB) – (E_proj – E_base) · AR=10% · RB=20%
6. sign_result(measurement) → measurement_id + signature
7. emit_event(result) → [TREE GROWTH MONITORING EVENT] → Edgar

4.3 Allometric species table (v1 seeds; needs field calibration)

Species	Formula source	Notes
Cacao (Theobroma cacao)	Chave et al. tropical / local Bahia equations	needs validation
Brazil nut	Chave tropical moist	—
Acai	palm-specific AGB	—
Mahogany	Chave tropical	—
Jatoba	Chave tropical	—

5. EVENT TYPES

- [TREE GROWTH MONITORING EVENT] — new event; per-measurement on-chain record: tree_id, dbh, agb, co2e, gps, captured_at, analysis_sha256, farmer_sig, worker_sig
- Reuses existing [TREE PLANTING EVENT] for the initial planting record
- Plan Vivo / VVB audit path: raw video (stored) + hash + analysis log = full provenance

6. IMPLEMENTATION ROADMAP (phased)

Phase	Deliverable	Effort	Owner	Exit criteria
P0 — Spec & contract	This spec; API contract frozen	0.5 d	Sophia	Gary sign-off
P1 — Browser capture	monitor_tree_growth.html in dapp_beta AND sunmint/monitor-tree-growth in truesight_me_beta: video capture, offline queue, signed POST	3–4 d	Dev (beta-first)	UAT on both beta sites
P2 — Python worker MVP	ingest → frames → card detect → DBH → CO■ → signed result	3–5 d	Sophia/Dev	Unit tests pass (compileall, ruff, pytest)

Phase	Deliverable	Effort	Owner	Exit criteria
P3 — Integration	worker endpoint ↔ dapp POST; response contract	2 d	Dev	End-to-end on beta
P4 — On-chain event	[TREE GROWTH MONITORING EVENT] in dao_protocol/Edgar	2 d	Dev	Event lands on ledger
P5 — Growth history view	per-tree measurements over time; farmer dashboard	2–3 d	Dev	History render on beta
P6 — Field calibration	species allometrics vs tape/calipers on pilot plots	ongoing	Field team	$R^2 \geq 0.9$
P7 — VVB audit prep	analysis reproducibility doc + video retention	1–2 d	Sophia	VVB-ready evidence pack

7. CHECKLIST (execution tracking)

- [ ] P0: spec signed off by Gary
- [ ] P1: monitor_tree_growth.html drafted (dapp_beta PR)
- [ ] P1: sunmint/monitor-tree-growth/index.html drafted (truesight_me_beta PR) — sunmint.truesight.me target
- [ ] P1: offline queue (IndexedDB) implemented
- [ ] P1: UAT on beta sites
- [ ] P2: worker scaffold in truesight_autopilot
- [ ] P2: extract_frames + detect_calibration_card
- [ ] P2: estimate_dbh + allometric_agb + carbon_math (PM002)
- [ ] P2: sign_result + emit_event
- [ ] P2: LOCAL TEST SUITE PASSES (compileall, ruff, ruff format, pytest)
- [ ] P3: endpoint wired to DApp POST
- [ ] P4: Edgar event type registered
- [ ] P5: growth history view on beta
- [ ] P6: field calibration data collected
- [ ] P7: VVB evidence pack assembled

8. RISKS & OPEN QUESTIONS

- Card detection in harsh light / occlusion → ARUCO markers more robust than plain contours; fallback to manual DBH entry + photo evidence (TREEO-style)

- Species allometrics for cacao are weak → P6 field calibration is critical; Plan Vivo allows project-specific equations
- Video size (50–200 MB) → compress to ≤ 720p / 10 fps in app before upload; worker cap 5 min video
- VVB acceptance of phone-video MRV → Plan Vivo participatory monitoring precedent (CommuniTree since 2010); keep raw video + hashes
- Worker deployment → autopilot box first; scale to dedicated worker if volume demands